

GANGASAGAR HL

Phone: 8105114611 Email: gangasagar.hi@iiitb.ac.in LinkedIn: linkedin.com/in/hlg-sagar-133013315 GitHub: github.com/Gangasagarhl

EDUCATION

M.Tech in Computer Science, International Institute of Information Technology Bangalore July 2024 – Present
GPA: 3.58/4

Bachelor of Engineering in Computer Science, Sapthagiri College Engineering August 2017 – July 2021
GPA: 3.59/4

EXPERIENCE

IoT Intern, SmartBridge, Remote [Link](#) Aug 2020

- Built end-to-end smart surveillance system using IBM Watson IoT for real-time monitoring and control.
- Integrated image capture with Cloud Object Storage, Cloudant DB, and Watson Visual Recognition for threat detection.
- Orchestrated flows via Node-RED and triggered Fast2SMS mobile alerts for unknown face detection.

PROJECTS

TeleManas Project (Ongoing) — Streamlit, LangChain, LoRA, MySQL, SQL [Link](#) Jan 2025 – Present

- Automated data analysis and generated visual insights to support government policy queries.
- Contributing to a government-funded academic initiative under the national TeleManas mental health program.
- Converted natural language queries into dynamic visualizations using LangChain and custom logic.

Heart Disease Detection — Airflow, MLFlow, Sklearn, AWS (ECR, EC2, S3), K8s, MongoDB Atlas [Link](#) May 2025

- Performed statistical analysis on patient data to identify risk factors and health trends.
- Built automated ETL pipelines with Airflow; logged and stored models using MLFlow and AWS S3.
- Implemented CI/CD with Docker, GitHub Actions, ECR, and EC2; simulated scalable K8s deployments locally.

Security System — Python, Flask, PyTorch, LangChain, TFLite, FAISS, Raspberry Pi 3B+ [Link](#) Apr 2025

- Deployed MobileNetV3-small (TFLite) on Raspberry Pi for real-time detection and alert-based media streaming.
- Integrated BLIP, T5, LLaMA 3.2 for alert summarization with Twilio/WhatsApp/Gmail notifications.
- Built FAISS-based RAG pipeline to index/summarize visual logs; exposed via REST API.

Transformers Arena — PyTorch, LoRA, Huggingface Hub, Qwen Transformer, Raspberry Pi 3B+ [Link](#) May 2025

- **GPT-2 Model:** Built custom GPT-2 in PyTorch for Raspberry Pi via hyperparameter tuning; published to PyPI as `sagar-neuron-gpt2`. Developed `sagar-neuron-gpt2`, a 38MB GPT-2 variant with perplexity 20 for edge inference; [Link](#).
- **One Word VQA:** Fine-tuned Qwen Vision Transformer using LoRA for one-word Visual QA on the Amazon-Berkeley dataset. [Link](#).

PUBLICATIONS

Predicting Soil pH using Mobile Captured Picture [Link](#) March 2021
Gangasagar HL, Jovin Dsouza, Bhagyashree B Yargal, Arun Kumar SV, Anuradha Badage. *International Research Journal of Engineering and Technology (IRJET)*, Vol. 08, Issue 07, July 2021.

TECHNICAL SKILLS

Languages: Python, C, C++ **Technologies/Frameworks:** TensorFlow, PyTorch, Docker, Git, GitHub, GitHub Actions, Flask, MySQL, SQL, ML Flow, Raspberry Pi

RELEVANT COURSEWORK

Software Systems Concrete Mathematics Machine Learning Mathematics For ML Enterprise System Development
Visual Recognition Multi Objective Machine Learning Project Elective OS

LEADERSHIP / EXTRACURRICULAR

Volunteered in institute-level activities and events under the Aikyam student body.

Spring 2020 – Present

District Level Table Tennis Winner.